

XJCO3011 — Coursework 2: Search Engine Tool

Submission Document

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Module: XJCO3011 — Web Services and Web Data
Submission Date: May 2026

(A) Video Demonstration Link

Platform: Google Drive / YouTube / OneDrive (public link required)

Video Link: [TO BE FILLED AFTER RECORDING]

The video must be accessible via a public URL. Test the link in incognito mode before submitting.

Video Requirements:

- Duration: Maximum 5 minutes (strict)
- Format: MP4 or MOV
- Content: Live demo + design explanation + testing + Git history + GenAI reflection
- Must cover all 5 components per coursework brief

(B) GitHub Repository URL

Repository: <https://github.com/Bartley007/xjco3011-cw2>

Repository Contents:

File / Directory	Description
.gitignore	Git ignore rules
requirements.txt	Python dependencies (requests, beautifulsoup4)
README.md	Project documentation with setup/usage instructions
src/	Core source code modules
crawler.py	Web crawler for quotes.toscrape.com (BeautifulSoup)
indexer.py	Inverted index builder (term frequency + positions)
search.py	Intersection-based multi-word search engine
main.py	Interactive CLI (build/load/find/print/exit)
tests/	Unit test suite
test_crawler.py	Crawler tests with mocked HTTP responses
test_indexer.py	Indexer tests (tokenization, build, save/load)
test_search.py	Search tests (single/multi-word, edge cases)
data/	Generated index files (index.json + docs.json)

Git Commit History (11 commits):

57fb037 | May 03 | fix: convert JSON string keys to int on load
c5d1bc9 | May 02 | docs: finalize for submission
d4b4661 | May 01 | docs: GenAI declaration
1b83cc3 | Apr 30 | feat: integrate all modules
70b4873 | Apr 29 | chore: update requirements
d134d18 | Apr 28 | docs: add README
c5e3054 | Apr 27 | test: add unit tests
e6eaf52 | Apr 23 | feat: implement CLI main loop
a643e75 | Apr 22 | feat: implement search engine
b7aadf2 | Apr 21 | feat: implement inverted indexer
9c213ae | Apr 20 | feat: implement web crawler

(C) Index File (Inverted Index)

File Name: **data/index.json** + **data/docs.json**

Description: Serialized inverted index generated by running the build command.

Index Structure:

Component	Format	Details
index.json	Dict[word → Dict[doc_id → List[pos]]]	Positional inverted index (100 quotes indexed)
docs.json	List[{text, author, tags, url}]	Full document storage from quotes.toscrape.com

Note: The index file will be attached separately on Minerva or included as a download link if too large.